

# Andrés Jauch

BSc Data Science Student · Basel / Dornach, Switzerland

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## EDUCATION

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**BSc Data Science** - FHNW University of Applied Sciences 2024 – Present

Brugg-Windisch, CH

- Focus on applied machine learning, statistical modeling, computer vision, and distributed systems.
- Core coursework: Linear Algebra, Probability & Statistics, Predictive Modeling, Software Engineering.

**Commercial Apprenticeship & Federal Matura (BM1)** - Wirtschaftsschule 2020 – 2023

Basel, CH

- Dual education combining business practice with advanced accounting, economics, and contract law.
- Graduated with Swiss Federal VET Diploma (EFZ) alongside academic baccalaureate (BM).

## SELECTED PROJECTS

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**Rhine Swimability Monte Carlo Simulation** | *Python, NumPy, SciPy, Pandas* GitHub ↗

- Processed 23 years of sensor time series to model swimmable summer conditions in the Basel Rhine.
- Implemented MLE parameter estimation, bivariate normal sampling with Cholesky decomposition ( $\rho \approx 0.50-0.57$ ), and a 10'000-run Monte Carlo simulation.
- Engineered a rolling-window ( $\pm 20$  days) continuous day-checker avoiding discrete month-boundary jumps.

**Tennis 3D Tracking & Analysis** | *Python, Computer Vision, Stereo Camera, ZED SDK, Streamlit* GitHub ↗

- Built an end-to-end stereo computer vision pipeline for real-time 3D ball and player position tracking.
- Calculated trajectory curves, court bounce detection, and velocity metrics served via an interactive dashboard.

**Brain Health Challenge** | *Python, Scikit-Learn, Random Forest, Jupyter* GitHub ↗

- Collaborative biomedical challenge predicting neurological clinical outcomes from clinical health indicators.
- Implemented and evaluated Random Forest classification baselines and feature importance pipelines.

## EXPERIENCE

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**Legal & Operations Assistant** - Advokatur & Rechtsberatung TRIAS AG 2023 – 2024

Basel, CH

- Structured and digitized physical case files into centralized, searchable document archives.
- Automated routine clerical workflows and coordinated deadline-sensitive legal documentation.

## TECHNICAL SKILLS

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- **Languages:** Python, SQL, R, TypeScript, JavaScript, HTML/CSS, Bash
- **Data Science & ML:** NumPy, SciPy, Pandas, Scikit-Learn, Matplotlib, Seaborn, Streamlit
- **Tools & Platforms:** Git, GitHub, Docker, Linux, Vite, PostgreSQL, Jupyter
- **Spoken Languages:** German (Native), Spanish (Native), English (Fluent)